

**THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS**

MSBA7014 Business Simulation

GENERAL INFORMATION
Instructor: Dr. Sundara Natarajan Panchanatham Email: sundar@hku.hk Office: Room 1325, K.K.Leung Building Consultation times: By appointment Pre-requisites: MSBA7002 Business Statistics and MSBA7003 Quantitative Analysis Methods Mutually exclusive: N/A Course Website: Moodle Reference books (for references only): • Banks, J., CARSON, J.S., Nelson, B.L., and Nicol, D.M., Discrete-Event System Simulation, 5th edition. Prentice-Hall, 2010. • Albright, S. Christian, and Wayne L. Winston. Business Analytics: Data Analysis & Decision Making. Cengage Learning, 2014. • W. David Kelton, Jeffrey S. Smith, David T. Sturrock. 2013. Simio and Simulation: Modeling, Analysis, Applications, 4th edition.
COURSE DESCRIPTION
This course introduces simulation modeling to analyze business processes and solve practical problems. Students will gain basic knowledge of Monte Carlo simulation and discrete event simulation and learn to build simulation models of business processes, interpret simulation results, and, more importantly, facilitate the decision-making process. This is a hands-on course on computer simulation with @Risk (www.palisade.com) and Arena. Note: We will use @Risk and Arena software. Both are Windows applications. You are required to have a Windows operating system installed on your laptop. If you use a Mac, kindly make sure you have access to a virtual machine with Windows or another device with the latest Windows OS.
COURSE OBJECTIVES
1. Develop complex simulation models of real or conceptual systems subject to risk. 2. Design an effective simulation experiment to be run on the model. 3. Analyze, interpret and communicate the simulation results. 4. Manage simulation projects, which includes input data analysis, simulation modeling, simulation experiment design, output analysis, and interpret results.
Programme Learning Outcomes

PLO1: Acquisition and internalization of knowledge of the programme discipline			
PLO2: Application and integration of knowledge			
PLO3: Inculcating professionalism and leadership			
PLO4: Developing global outlook			
PLO5: Mastering communication skills			
COURSE LEARNING OUTCOMES			
Course Learning Outcomes		Aligned Programme Learning Outcomes	
CLO1 Understand the concept of Monte Carlo simulation and be able to apply it in risk analysis with spreadsheet simulation.		CLO1 - PLO1, 2	
CLO2 Understand the concept of discrete-event simulation and be able to develop discrete-event simulation models with Simio.		CLO2 - PLO1, 2	
CLO3 Master important analytics skills for a simulation project, which includes input analysis, output analysis, and comparisons of alternative systems.		CLO3 - PLO1, 2	
CLO4 Appropriately design simulation experiments for decision-making.		CLO4 - PLO1, 2, 3, 4	
CLO5 Be able to interpret and communicate results from simulation models.		CLO5 - PLO1, 2, 3, 4, 5	
COURSE TEACHING AND LEARNING ACTIVITIES			
Course Teaching and Learning Activities		Expected contact hour	Study Load (% of study)
T&L1. Lectures with interactive presentation		36	30%
T&L2. Skill-Building Exercise		30	25%
T&L3. Assignments and Practice Problems		30	25%
T&L4. Project		24	20%
Total		120	100%
Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. Written assignments	Four assignments will be assigned throughout the semester	40%	CLO1, 2, 3, 5
A2. In-class participation	Attendance and discussions	20%	CLO1, 2, 3, 5
A3. Group projects	One final course project	40%	CLO2, 3, 4, 5
	Total	100%	
STANDARDS FOR ASSESSMENT			
Course Grade Descriptors			
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Excellent writing report		

B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Good participation in discussions • Good writing report
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Some participation in discussions • Acceptable writing report
D+, D	• Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Minimal or no participation in discussions • Marginally acceptable writing report
F	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Minimal or no participation in discussions • Poorly writing report

Assessment Rubrics for Each Assessment

Written group project and exams

A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills • Excellent writing report
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills • Good writing report
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills • Acceptable writing report
D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills • Marginally acceptable writing report
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills • Poorly writing report

Class and tutorial participations

A+, A, A-	• High participation in discussions • Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	• Good participation in discussions • Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook

	• Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	• Some participation in discussions • Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	• Minimal or no participation in discussions • Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Respect others and follow the class rules (no chatting and do not use cell phone)
F	• Minimal or no participation in discussions • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)
COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE	
Lecture1	1 Introduction and Review of Probability and Decision Theory
Lecture2	2 Monte Carlo Simulation with Excel
Lecture3	3. Simulation with @Risk
Lecture4	4. Input Analysis
Lecture5	5. @Risk Applications
Lecture6	6 Queueing models and Discrete-Event Simulation
Lecture7	7 Simulation with Simio
Lecture8	8 Output analysis
Lecture9	9 Decision Analysis with Simulation
Lecture10	10 Simulation Analytics
REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)	
Lecture notes will be uploaded to course website.	
MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE	
X Conducting mid-term survey in additional to SETL around the end of the semester ○ Online response via Moodle site ○ Others: _____ (please specify)	
COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)	
An orderly learning environment is extremely important for this course. Disruptive behaviors are inconsiderate to other students as well as to the instructor, and are absolutely unacceptable. Talking during lectures, arriving to class late, and any other disruptions of mobile devices are not allowed; students who are responsible for any of these actions will be subject to academic penalty and will be asked to leave the classroom. Any dishonesty - such as cheating, false representation, plagiarism, etc.—that comes to my attention will result in an F in the course. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Scholastic dishonesty also includes but is not limited to, providing false or misleading information to receive a postponement or an extension on an exam or other assignment. The responsibilities of both students and faculty with regard to scholastic dishonesty are described in detail in the Disciplinary Committee Regulations (http://www.hku.hk/pubunit/cal99/104f.htm). By teaching this	

course, I have agreed to observe all of the faculty responsibilities described in that document. By enrolling in this class, you have agreed to observe all of the student responsibilities described in that document. If the application of that policy statement to this class and its assignments is unclear in any way, it is your responsibility to ask me for clarification.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

We will use @Risk and Arena software. Both are Windows applications. You are required to have a Windows operating system installed on your laptop. If you use a Mac, kindly make sure you have access to a virtual machine with Windows or another device with the latest Windows OS.

Tentative Outline of Lectures:

Lecture 1	Introduction and Review
Lecture 2	Monte Carlo Simulation with Excel
Lecture 3	Introduction to @Risk
Lecture 4	Applications with @Risk
Lecture 5	Applications with @Risk
Lecture 6	Introduction to Arena (Expert)
Lecture 7	Introduction to Arena (Expert)
Lecture 8	Simulation Input Analysis
Lecture 9	Simulation Output Analysis
Lecture 10	Presentation